Project Tenacity: A Sketch, Simulate and Validate Planner for Extravehicular Walks on Mars, Modelled on Rover Drive Planning

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Live tool: yara-hue24.github.io/Project-Tenacity · Source code (MIT): github.com/yara-hue24/Project-Tenacity · Demo slides

Abstract

Rover drives on Mars are never sent unchecked: planners at the Jet Propulsion Laboratory sketch each drive on the terrain, simulate it and validate it against rules before uplink. No equivalent open tool exists for a crew on foot, for whom the binding constraint is not distance but the time needed to regain shelter after a suit failure. We present Tenacity, a browser-based planner that applies the sketch, simulate and validate loop to extravehicular walks at Gale crater. A walk is laid out on orbital imagery or found by a least-time path search over a 6 m elevation grid derived from the HiRISE 1 m model. A step-by-step simulation predicts duration, metabolic effort, oxygen, carbon dioxide, cooling water and battery use, and the emergency return time from every point of the route. Fifteen flight rules, each traced to a NASA document or declared as our own, yield a single verdict (GO, NO-GO, PROVISIONAL or NOT EVALUATED) and name the rule that decides. More than 25 data layers feed the same rules that are drawn on the map. Every constant carries a provenance tag. Two reference walks are reported: one accepted with a 47 minute margin, one rejected because the farthest point lies 62 minutes from the refuge. The tool is intended for astronauts and mission operations personnel, and states its limits explicitly.

Keywords: extravehicular activity; traverse planning; Mars; Gale crater; flight rules; terrain analysis; least-time path; mission operations software.

1. INTRODUCTION

A map that shows where things are cannot answer the question a surface crew asks before opening the airlock: can we go, and can we get home? On Mars the limit of a walk is set by the suit. The backup life support of NASA's exploration suit is designed to bring a crew member back to a safe haven within 60 minutes [4], and Artemis III bounds walking distance in the same way [5]. A route that looks short in plan view may be far in time once slope, loose ground and the detours around steep terrain are counted.

Robotic missions already solve the corresponding problem. Every drive of Curiosity and Perseverance is planned in the Rover Sequencing and Visualization Program (RSVP), in which drive legs are sketched on a terrain model, simulated, and checked by flight rules before the sequence is sent [1, 2]. Tenacity transfers that working method to a walking crew.

Intended users. Tenacity is a professional planning tool, not a public outreach product. It is designed for two groups: astronauts, who plan the walk and then follow it, and mission operations personnel (flight controllers, EVA officers and planners), who must clear the plan before egress. This choice explains the dense, workstation-style interface and the use of flight-rule and alert conventions.

Contributions. (i) A plan model and step-by-step walk simulation for suited traverses, with all constants sourced and tagged; (ii) a terrain engine running entirely in the browser that computes slope, cast shadow, line of sight, an emergency walk-home time field and least-time paths; (iii) a set of fifteen flight rules with an explicit treatment of missing or stale data; (iv) an integration of more than 25 NASA and partner data layers in which the layers drawn on the map are the same data the rules evaluate; and (v) an explanation view that shows why a path was chosen.

2. BACKGROUND AND RELATED WORK

Rover drive planning. RSVP consists of cooperating tools: MobSketch for sketching drive legs, RoSE for editing the command sequence, SSim for simulation, HyperDrive for terrain visualisation and playback, and ROCS for time, shadow and path services [1, 2]. Table 1 gives the correspondence we adopted. The comparison is ours, drawn from JPL's public description of the suite and from figures in Rankin et al. [2].

Traverse planning for crews. SEXTANT, developed at MIT with NASA, computes least-cost traverses for suited explorers from an elevation model, using a walking-speed function of slope credited to Marquez and the load-carriage model of Santee et al. for metabolic cost [3, 10, 11]. Tenacity re-implements both functions and scales them to NASA design values (Section 4).

Suit and life-support constraints. Capacities and rates are taken from the description of the xEMU portable life support system [4], the Artemis III surface requirements [5], NASA's Life Support Baseline Values and Assumptions Document [6] and the Apollo 15 planning memorandum on EVA capability [7].

Crew displays. Alert classes and colours follow NASA-STD-3001 Volume 2 [8]; layout guidance was taken from NASA's crew-interface reference material [9].

Table 1. Components of JPL's rover planning suite and their counterparts in Tenacity.

Rover planning (RSVP)	Tenacity, for a walk
MobSketch: drive legs drawn on the map, activity list	Legs and stations on the map, with an activity list
RoSE: command sequence table and status console	Sequence table (one row per walk, work or return step) and a console
SSim: predicts the effect of a sequence before uplink	Walk simulation: time, oxygen, CO_2, cooling water, battery, walk-home time
HyperDrive: terrain views, clocks, playback	Terrain layers, clock strip (UTC, Mars mean and true solar time, elapsed time), playback, 3D flyover
ROCS: time, shadow, path and terrain services	Mars time, Sun position, cast shadow, line of sight and path finding, computed in the browser

3. DATA

The study area is the part of Gale crater explored by Curiosity, from the landing site to the boxwork terrain on lower Mount Sharp. Table 2 lists the datasets. The rover's own track and stop positions define the "proven ground" of the planner: legs that follow it are drawn solid, legs on ground known only from orbit are dashed and flagged by rule FR-05.

The HiRISE elevation model was resampled to 1552 × 3200 cells of about 6.1 m and stored as 16-bit integers with 0.1 m resolution (5 million cells, delivered compressed). CRISM mineral-indicator maps were shifted by 150 to 250 m to register them to the base map; a residual error of 30 to 60 m remains. In-situ measurements (CheMin, APXS, ChemCam, DAN) are attached to the rover stop at which they were acquired. All derived files are produced by scripts in the repository from public services and can be regenerated.

Table 2. Datasets used. Resolution is that of the product as used in the tool.

Dataset	Resolution or size	Use in Tenacity	Ref.
Curiosity location map (NASA/JPL-Caltech): track and stops	38.8 km; 1,388 stops, sols 3 to 5070	Proven ground; station positions; base-map tiles (HiRISE 25 cm, CTX 6 m)	[13]
MRO HiRISE elevation model of Gale (NASA Mars Trek)	1 m, resampled to 6.1 m	Slope, roughness, relief lines, shadow, line of sight, walk-home time, path finding, 3D terrain	[14]
Curiosity raw images (Navcam)	One frame per stop	View on arrival at a station	[13]
CheMin, APXS, ChemCam (NASA PDS Geosciences Node)	48 samples; 1,784 spots; 4,393 targets	Mineralogy and chemistry at the point probed	[15]
MRO CRISM mineral-indicator maps (NASA PDS)	18 m	Hydrated minerals, clays, iron minerals from orbit	[16]
Mars Odyssey THEMIS thermal inertia and night infrared (ASU JMARS)	100 m	Loose-ground indicator (rule FR-11)	[17]
DAN subsurface water along the traverse (Roscosmos instrument)	To sol 3333	Water-equivalent hydrogen at the point probed	[18]
NASA DONKI, WSA-Enlil simulations	Event forecasts	Solar eruptions forecast to reach Mars (rule FR-10)	[12]
Curiosity's proposed path up Mount Sharp; USGS SIM 3292	13 regions; 1:20,000,000	Named science regions; global geologic unit	[19, 20]

4. METHODS

4.1 Plan model

A plan consists of a refuge (a habitat or a pressurised rover that can be re-parked), an ordered list of stations, and the legs between them. Each leg is walked in one of three ways: along the rover's track, in a straight line, or on the least-time path found by the terrain engine under the slope limit. Each station carries a work duration and a work intensity. The plan is expanded into a list of steps (walk, work, return) that the simulation consumes; the same list is shown to the user as a sequence table.

4.2 Walking speed

Speed over the map is a function of the slope s (degrees, positive uphill) along the direction of travel. We use the piecewise-linear lunar curve implemented in SEXTANT [3, 10]:

$$v_0(s) = \begin{cases} 0.05 & s \le -20 \\ 0.095\ s + 1.95 & -20 < s \le -10 \\ 0.06\ s + 1.6 & -10 < s \le 0 \\ -0.2\ s + 1.6 & 0 < s \le 6 \\ \max(0.05,\ -0.039\ s + 0.634) & 6 < s \le 15 \\ 0.05 & s > 15 \end{cases} \quad (1)$$

with v_0 in m/s. The curve gives 1.6 m/s on level ground. It is scaled so that the level speed equals the planned speed V: $v(s) = v_0(s) \cdot V \,/\, 1.6$. The default $V = 3.3$ km/h is the Apollo 15 planning value for walking traverses [7]. Slope is measured over a 20 m baseline along the path. For the emergency return the same shape is scaled to the emergency pace.

4.3 Metabolic power

Metabolic power follows the load-carriage model of Santee et al. [11] as coded in SEXTANT, with total mass m (crew member 82 kg plus suit 121 kg), slope angle α and Mars gravity $g = 3.71$ m/s²:

$$P = \max\{\ P_{\min},\ k\,(3.28\ m + 71.1)(0.661\ v\cos\alpha + 0.115) + C(\alpha)\ \} \quad (2)$$

where $C = 3.5\ m\ g\ v \sin\alpha$ uphill and $C = 2.4\ m\ g\ v \sin\alpha \cdot 0.3^{|\alpha|/7.65^\circ}$ downhill. The original model describes unsuited walking on Earth and returns about 530 W for a suited mass on level ground; the factor k rescales the level term so that level walking at V costs the xEMU design rate of 352 W [6]. Because the speed curve slows climbs strongly, the raw formula would make a climb cheaper than level walking; since Apollo crews recorded their highest rates on climbs, uphill power is floored at the level value. $P_{\min} = 144$ W is the lowest suited rate on record [6]. The emergency return is charged at a flat 422 W [7]. Figure 1 shows both functions.

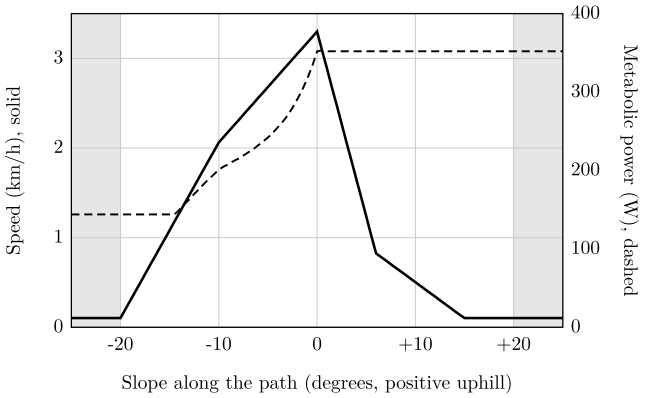


Figure 1. Walking speed (solid) and metabolic power (dashed) against slope, as used by the simulation at the default planned speed of 3.3 km/h. Shaded bands lie beyond the 20° suit limit. Both curves are extrapolations to Mars and are tagged ASSUMED.

4.4 Consumables

For metabolic power P (W), consumption rates are

$$\begin{aligned} \dot{m}_{O_2} &= 2.56{\times}10^{-4}\ P + 0.005 \quad \text{kg/h} \\ \dot{m}_{CO_2} &= 1.2375 \cdot (2.56{\times}10^{-4}\ P) \quad \text{kg/h} \\ \dot{m}_{H_2O} &= (P + 92.9)\ /\ (2.45{\times}10^{6}) \quad \text{kg/s} \end{aligned} \quad (3)$$

and the battery is drained at a constant 80 W. The oxygen coefficient is derived from 0.075 kg per crew-member-hour at 293 W, the constant term is suit leakage, the carbon dioxide factor is a respiratory quotient of 0.90 times the molar mass ratio, and cooling water is evaporated to reject metabolic heat plus 92.9 W of other loads [6]. At 352 W this gives 0.095 kg/h of oxygen and 0.65 kg/h of water. Starting quantities are 0.771 kg of oxygen, 3.856 kg of usable water and 756 Wh [4]. Table 3 lists the principal constants with their provenance tags.

Table 3. Principal constants of the simulation. Tags: NASA VALUE, taken from a NASA document; DERIVED, computed here from such values; ASSUMED, a choice of ours.

Quantity	Value	Tag	Source and remark
Usable oxygen at egress	0.771 kg	NASA VALUE	xEMU primary oxygen vessel, 1.7 lbm usable [4]
Usable cooling water	3.856 kg	DERIVED	8.5 lb of the 10 lb feedwater assembly, read from the xEMU schematic [4]
Battery energy; electrical load	756 Wh; 80 W	NASA VALUE; ASSUMED	xEMU primary battery [4]; no published power budget was found
Level walking: speed; power	3.3 km/h; 352 W	ASSUMED; NASA VALUE	Apollo 15 planning speed [7]; xEMU average design metabolic rate [6]
Emergency return: time; power	60 min; 422 W	NASA VALUE	xEMU auxiliary loop abort capability [4, 5]; Apollo walk-back rate with 20% pad [7]
Consumables reserve	30 min	NASA VALUE	Apollo 15 planning memorandum [7]
Slope limit; impassable slope	20°; 30°	NASA VALUE; ASSUMED	Artemis suit requirement "up, down, across a 20-degree slope" [5]; SEXTANT masks terrain above 35° [3]
Maximum duration	8 h	NASA VALUE	Suit requirement [5]
Slow-pace factor	1.375	DERIVED	Planned 3.3 km/h over the 2.4 km/h average of an Apollo walking traverse [22]
Work rates: light; moderate; heavy	144; 286; 530 W	NASA VALUE	Apollo surface minimum and average [6]; Mars PLSS sizing maximum [21]

4.5 Terrain engine

All terrain products are computed in the browser, in a web worker, from the elevation grid.

Slope, roughness and shading. Slope and aspect are obtained from central differences, which average over about 12 m. Roughness is the local spread of elevation. Hill shading and cast shadows are computed for the Sun's azimuth and elevation at the plan time; a cell is in cast shadow when it lies below the shadow edge thrown by ground up-sun of it.

Line of sight. A viewshed from the refuge assumes a 3 m antenna and a 1.8 m helmet height and lowers distant ground by $d^2/2R$ for planetary curvature.

Time fields and paths. Travel time is accumulated with Dijkstra's algorithm over the eight neighbours of each cell. A move covers the horizontal distance d between cell centres; its slope is that of the move itself, in the direction walked, so the cost is anisotropic: $t = d \,/\, v(s)$. Cells steeper than the impassable slope, and cells inside keep-out zones drawn by the user, cannot be entered. Three products follow: (a) the least-time path of a leg; (b) the walk-home field, the time from every cell *to* the refuge at emergency pace; and (c) the 60 minute contour of that field, drawn on the map as the suit-failure return limit.

Explanation. For any leg found by the engine the tool reports the length of the straight line, the length of that line lying on ground over the limit, and the length, climb, drop and duration of the chosen path; it overlays impassable ground in red, slow ground in yellow and the rejected straight line (Figure 3, left).

4.6 Time and the Sun

Mars local mean and true solar time, areocentric solar longitude, and the Sun's azimuth and elevation at the refuge are computed with the Mars24 algorithm [23]. Sunrise and sunset assume a flat horizon.

4.7 Flight rules and verdict

Fifteen rules are evaluated along the whole walk (Table 4). Each returns one of five levels: pass, advisory, caution, warning, or *not evaluated*. The last is central to the design: a rule whose data are missing or older than their validity (for example a space-weather forecast older than 48 hours) is never counted as a pass.

The verdict is NO-GO if any rule raises a warning. Otherwise it is PROVISIONAL while the terrain engine or forecast data are still loading, NOT EVALUATED if any of the three critical rules (FR-01 return time, FR-02 oxygen, FR-04 slope) could not be evaluated, and GO in the remaining case. The interface always shows the verdict, the margin, and the rule that limits it.

FR-04 uses the larger of the slope along the path and the slope of the ground itself, so that a traverse across a steep face is caught. FR-14 repeats every timed rule with walking 1.375 times slower, and FR-15 flags walks that spend long on slopes where the speed model is at its weakest.

Table 4. Flight rules. Rules without a NASA basis are declared as our own.

Rule	Condition checked	Basis
FR-01	At every moment the refuge is within 60 min at emergency pace	xEMU emergency return capability [4]; Artemis III [5]
FR-02	Oxygen left covers the emergency walk home plus a 30 min reserve	Apollo 15 planning memorandum [7]
FR-03	Return before sunset, with the reserve	Mars24 for the season at the refuge [23]
FR-04	Slope within 20° up, down or across, at every point	Artemis suit requirement [5]; HiRISE elevation [14]
FR-05	Legs on ground not driven by a rover are flagged	Our rule
FR-06	Line of sight to the refuge	Computed from the elevation grid
FR-07	Duration no longer than 8 h	Suit requirement [5]
FR-08, 09	Battery and cooling water cover the walk home plus reserve	xEMU capacities [4]; battery load assumed
FR-10	No solar eruption forecast to reach Mars within 2 days; glancing impact is a caution; forecasts older than 48 h are not evaluated	NASA DONKI WSA-Enlil [12]
FR-11	Loose ground on the route	THEMIS thermal inertia [17]
FR-12	Route stays out of keep-out zones	Zones set by the crew
FR-13	At least two crew on the walk	NASA plans surface walks in pairs
FR-14	Timed rules still hold when walking is 1.375 times slower	Apollo traverse average [22]
FR-15	Flags long stretches where the modelled pace falls below 0.5 km/h	Our rule, on the model's own limits

4.8 Provenance

Every constant is stored with its value, unit, source and one of four tags: MEASURED, NASA VALUE, DERIVED or ASSUMED. The tags are shown in the interface next to the numbers they qualify and listed in full in a Sources tab, so that a reviewer can see at once which results rest on measurements and which on assumptions.

4.9 Interface

The interface is organised as a workstation (Figure 2): the plan as an activity list, the map, the simulation and rules, a clock strip with playback, and the sequence table with a console. Docks can be resized, folded, hidden and swapped; plans, notes and the history of changes persist in the browser. Alert colours have fixed meanings following NASA-STD-3001 [8] (red warning, yellow caution, blue advisory) and are never the only cue: each alert also carries a class letter. Routes are drawn as rover planners draw drives, with named legs, stations with an arrival circle and keep-out zones. A probe reads every layer at a clicked point, and a 3D flyover drapes HiRISE imagery on the elevation model and shows Curiosity's photograph of each station (Figure 3, right). A guided tour introduces the screen to new users.

4.10 Implementation and verification

Tenacity is a static web page written in HTML, CSS and JavaScript modules, with the Leaflet library for the 2D map and our own WebGL code for the 3D view. There is no server and no build step. The simulation, the Mars clock, the terrain engine and the 3D mathematics are covered by 113 automated tests. In addition, the tool and its data were subjected to an adversarial review written from the standpoint of mission operations personnel; the defects it found (among them cross-slope handling, stale forecast handling and an optimistic verdict while data were loading) were corrected before release.

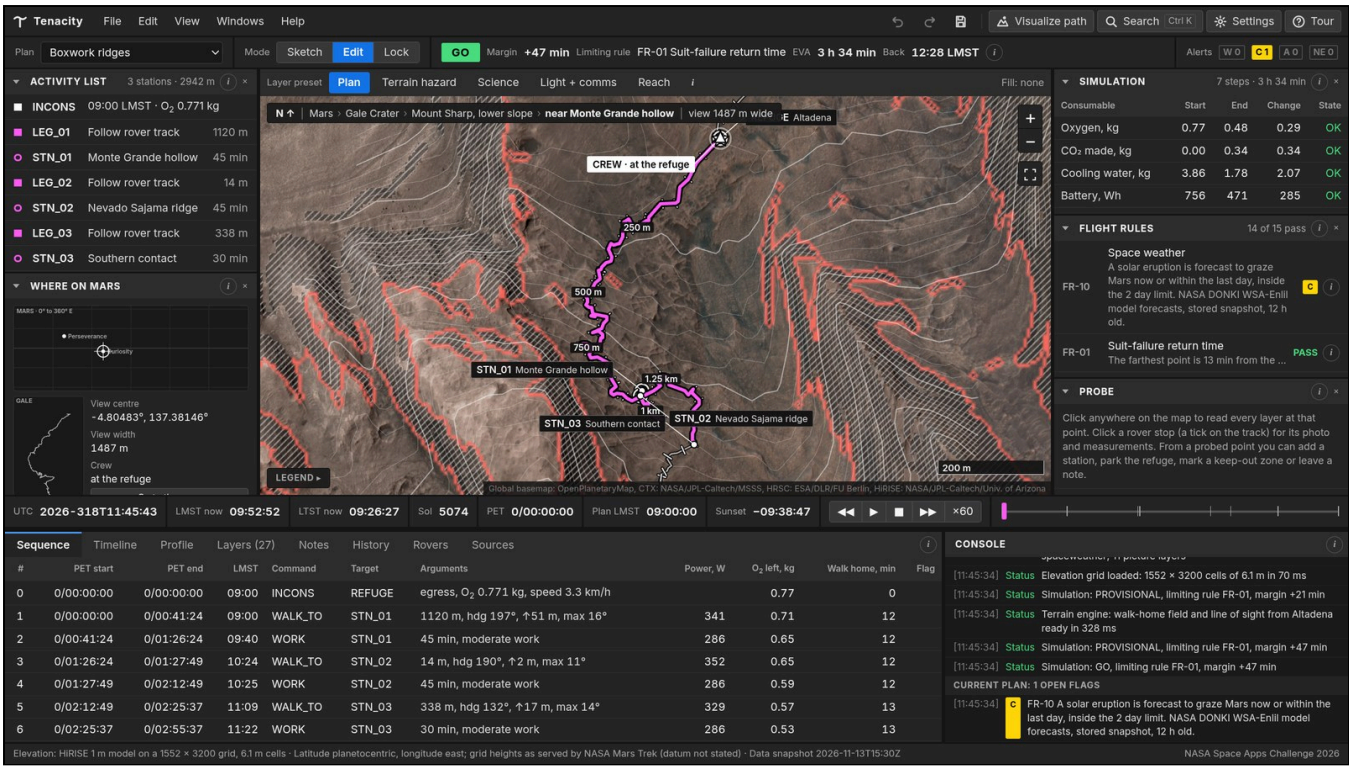


Figure 2. The planner with reference walk A. Left: the plan (initial conditions, legs, stations). Centre: the map with the route in magenta, ground over the slope limit hatched, the return limit in red and the crew marker. Right: simulation, flight rules and probe. Below: Mars clocks, playback and the sequence table. Imagery: NASA/JPL-Caltech/Univ. of Arizona.

5. RESULTS

5.1 Reference walks

Two ready-made walks on ground driven by Curiosity illustrate the two outcomes (Table 5). Both were run with default parameters and two crew.

Walk A, "Boxwork ridges". Three stations on 2.94 km of rover track, with 2 h of work. The simulation gives 3 h 34 min. The farthest point is 13 min from the refuge at emergency pace against the 60 min limit, so FR-01 is the limiting rule with a margin of 47 min. The steepest ground is 15.8°, the refuge stays in sight, and the slow-pace case still holds with 82 min to spare. Fourteen of fifteen rules pass. The exception was FR-10, a caution on the day of the run, because a solar eruption was forecast to graze Mars. Verdict: GO.

Walk B, "Dingo Gap to Pahrump Hills". A long traverse of 9.91 km along the rover track. Consumables at nominal pace are sufficient (0.40 kg of oxygen left) and no slope exceeds 12.4°, so a check of distance, slope and oxygen alone would accept it. The tool rejects it: the farthest point is 62 min from the refuge at emergency pace (FR-01), and in the slow-pace case the cooling-water rule fails (FR-14). The refuge is also out of sight for 8.8 km of the route (advisory). Verdict: NO-GO.

Table 5. Simulation results for the two reference walks (run on 14 November 2026; FR-10 depends on the forecast of the day).

Quantity	Walk A: Boxwork ridges	Walk B: Dingo Gap to Pahrump Hills
Route length	2.94 km	9.91 km
Duration	3 h 34 min	4 h 14 min
Oxygen used (of 0.77 kg)	0.29 kg	0.37 kg
Cooling water used (of 3.86 kg)	2.07 kg	2.59 kg
Battery used (of 756 Wh)	285 Wh	338 Wh
Farthest point from refuge, emergency pace	13 min	62 min
Steepest ground on route	15.8°	12.4°
Slow-pace case (× 1.375)	4 h 09 min, all timed rules hold	5 h 39 min, cooling-water rule fails
Rules passed	14 of 15	11 of 15 (2 warnings, 1 caution, 1 advisory)
Verdict; limiting rule; margin	GO; FR-01; +47 min	NO-GO; FR-01 and FR-14; −2 min

5.2 Path finding off the rover track

Adding a station by a single click 483 m in a straight line from the previous one, on ground known only from orbit, produced the case in Figure 3 (left). The straight line crosses 139 m of ground steeper than 20°. The path found is 1092 m long, climbs 39 m and drops 40 m, stays at or below the limit and takes 34 min: more than twice the straight distance, for a route that is permitted. The leg is flagged by FR-05 and its steepest ground raises a caution under FR-04, which the user can see and act on.

5.3 Performance

On a laptop, in a browser without hardware graphics acceleration, decoding the 5-million-cell grid took about 0.1 s and the walk-home field with the line-of-sight map about 0.3 s. Path finding for a leg completes within a few seconds. The interface is usable while these products load, during which the verdict reads PROVISIONAL.

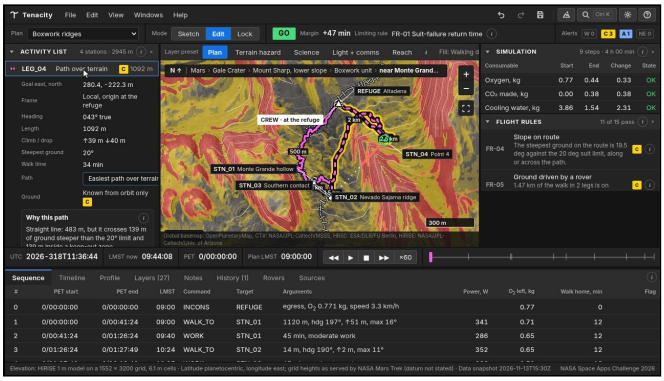


Figure 3 (left). "Why this path" for the leg of Section 5.2: ground that cannot be crossed in red, slow ground in yellow, the rejected straight line dashed.

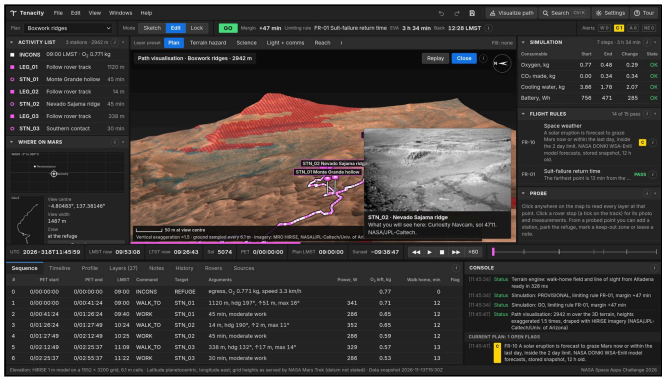


Figure 3 (right). 3D flyover: HiRISE imagery on HiRISE elevation, with Curiosity's Navcam photograph of the station.

6. DISCUSSION AND LIMITATIONS

The two walks show the point of the approach. Walk B is acceptable by distance, slope and oxygen, and is rejected by a rule that only a time field over real terrain can evaluate. The explanation view and the rule text make the reason visible, which is what a controller clearing the plan needs.

The results are bounded by the following limits, which the tool also states.

- **The walking model is unvalidated for Mars.** Suited walking speed and effort in Mars gravity have never been measured. The speed curve comes from lunar data and makes climbs very slow (about 0.5 km/h at +10° and 0.1 km/h near +20°); the power model is an Earth treadmill model rescaled to a design rate. Both are tagged ASSUMED, and FR-14 and FR-15 exist to expose plans that depend on them.
- **Terrain.** The grid is suited to relative height. It lies a few metres to about 20 m from the rover's own elevations, and its vertical datum is not stated by the service. Slope averaged over 12 m understates short steps. Paths turn in 45° steps and read up to 8% long. Rocks and sand at footstep scale are not seen by any layer.
- **Navigation.** The crew's position is assumed known; position error is not modelled.
- **Time.** A plan has no date of its own: Sun, season and space weather are those of the day the tool is opened.
- **Environment.** Space weather is a model forecast, not a measurement at Mars; when the service cannot be reached and the stored copy is older than 48 hours the rule reads not evaluated. Dust storms, thermal loads and particle dose are not modelled.
- **Emergency case.** The return assumes an able walker; an injured crew member is not modelled.
- **Status.** Tenacity is a prototype for planning and training, tested in desktop browsers. It is not a certified flight product.

7. FUTURE WORK

Planned extensions, none of which is built: a date and time for each plan; tracking of the crew's real position during the walk; a second site at Jezero with Perseverance's data; replacement of the walking model when measured suited-walking data exist; a rover-assisted mode in which the rover drives to meet the crew; and a light theme for bright cabins.

8. CONCLUSION

The sketch, simulate and validate loop that protects rover drives can be given to a crew on foot with public NASA data and a web browser. Tenacity answers one question, whether a walk can be made and the refuge regained, names the rule that decides, shows where on the map it applies, and is explicit about which of its numbers are measured and which are assumed.

AVAILABILITY

The tool runs at the live link above in a desktop browser (Chrome, Edge or Firefox) with no login or installation; an internet connection is needed for NASA's map imagery. A guided tour is offered on the first visit. Walk A loads by itself; walk B is selected in the Plan list. Switching Mode to Sketch and clicking the map reproduces Section 5.2. The source code, data-building scripts and tests are in the repository under the MIT licence.

USE OF ARTIFICIAL INTELLIGENCE

The team chose the challenge, set the concept (a rover-planner-style tool for astronauts), the requirements and the design direction, and reviewed and tested the result. Research was carried out in two ways. Part of it was done by the team by hand: finding and reading primary sources, among them JPL's description of the rover planning tools, NASA's crew-interface material, NASA's map of Curiosity's proposed path up Mount Sharp, NASA Mars Trek, the USGS geologic map and the rover image archives. The team also used Perplexity, an AI search assistant, to locate and cross-check sources. Further background research, most of the program code (JavaScript, CSS, HTML) and the Python scripts that build the data files were produced with Claude, an AI coding assistant by Anthropic, working under the team's direction; an AI-run adversarial review was used to find defects; and this document and the demo slides were drafted with the assistant and checked by the team. No AI-generated images, video or audio are part of the project: all pictures are screenshots of the working tool or NASA imagery inside it.

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